

Javad Noorbakhsh

Bioinformatics & Machine Learning Scientist

javad.noorbakhsh@gmail.com | Boston, MA | javadnoorb.github.io | linkedin.com/in/javadnoorbakhsh | Google Scholar

SUMMARY

Computational biologist working on AI methods for cancer spatial omics and computational pathology, with a research background spanning cancer genomics and systems biology. Leads technical teams and cross-functional research collaborations.

EXPERIENCE

Bioinformatics & Machine Learning Scientist

Jul 2024 – Present

Jackson Laboratory for Genomic Medicine, Farmington, CT

- Applied pathology foundation models to accurately identify and classify blood cell types within Giemsa-stained tissue slides.
- Engineered ML algorithms to cross-evaluate and compare spatial omics profiles between human and mouse models.
- Benchmarked the impact of H&E staining variations to optimize foundation model diagnostic performance.
- Evaluated spatial transcriptomics data from cancer patient cohorts to quantify therapeutic effects on microenvironment composition.
- Formulated mathematical models of tumor evolution and subclonal dynamics, enhancing precision in treatment strategy selection.
- Co-authored peer-reviewed reviews and competitive grant proposals advancing AI applications in spatial omics.

Principal Scientist

Apr 2022 – Feb 2024

Kojin Therapeutics, Boston, MA

- Spearheaded AI-driven multi-omics analysis, directly enabling fundraising and a pharmaceutical partnership.
- Established high-throughput assay screening pipelines, identifying novel small-molecule hits for drug development.
- Deployed graph neural networks (GNNs) for molecular design, accelerating target discovery and experimental drug candidate validation.
- Architected attention-based multi-instance learning pipelines on histology images, uncovering potential drug-sensitive cancer subtypes.

Lead Scientist

Jan 2020 – Apr 2022

Broad Institute of MIT and Harvard, Cambridge, MA

- Orchestrated the Cancer Dependency Map omics team, managing end-to-end RNAseq, WES, and WGS pipelines to deliver quarterly releases across public, institutional, and commercial portals.
- Standardized and automated production pipelines across major projects, achieving reproducible, timely data distribution.
- Directed ML initiatives aligning patient tumor NGS data with cell lines, boosting alignment accuracy and tool reliability.

Research Scientist

Oct 2017 – Dec 2019

Jackson Laboratory for Genomic Medicine, Farmington, CT

- Pioneered a cloud-based deep learning framework for tumor state classification across histopathology images, attaining AUCs > 0.9 across 19 cancer types.

- Guided the development and deployment of an image processing framework for 3D organoid segmentation, producing an open-source software package.

Postdoctoral Fellow

Feb 2015 – Oct 2017

Jackson Laboratory for Genomic Medicine, Farmington, CT

- Investigated copy number variations and mutation calling dynamics in cancer, identifying key biomarker correlations with patient survival.
- Constructed mathematical models of clonal evolution, improving quantitative assessment of genetic heterogeneity across diverse tumor types.

EDUCATION

Ph.D., Physics — Boston University, Boston, MA

Sep 2009 – Dec 2014

M.Sc., Physics — Sharif University of Technology, Tehran, Iran

Sep 2006 – Dec 2008

B.Sc., Electrical Engineering — Amirkabir University of Technology, Tehran, Iran

Sep 2001 – Dec 2005

SKILLS

Quantitative Analysis Machine Learning, Deep Learning, Bioinformatics, Statistics, Image Processing, Mathematical Modeling, Cheminformatics

Programming Python, TensorFlow/PyTorch, Scikit-learn, R, Bash, HPC, Cloud Computing, Containerization, Kubernetes, Version Control, C++, MATLAB

Leadership People Management, Mentorship, Team Collaboration

PUBLICATIONS

Full record on [Google Scholar](#).

1. [Prediction of outcome from spatial Protein profiling of triple-negative breast cancers](#). Ali Foroughi Pour, Te-Chia Wu, **Javad Noorbakhsh**, Jan Martinek, Santhosh Sivajothi, et al. *Communications Medicine*, 2026.
2. [RANKL inhibition spatially rewires the microenvironment of luminal breast cancer](#). Mario Rodriguez-del Collado, Andrea Vethencourt, Alexandra Barranco, Jaime Martinez de Villarreal, **Javad Noorbakhsh**, et al. *medRxiv*, 2026.
3. [DIANNE: Segmentation-Free Localization of Histology Differential Attributes](#). Sergii Domanskyi, Jill C Rubinstein, Todd B Sheridan, Adam Thiesen, **Javad Noorbakhsh**, et al. *bioRxiv*, 2026.
4. [Emerging AI approaches for cancer spatial omics](#). **Javad Noorbakhsh**, Ali Foroughi Pour, Jeffrey Chuang. *GigaScience*, 2025.
5. [Germline variation contributes to false negatives in CRISPR-based experiments with varying burden across ancestries](#). Sean A Misek, Aaron Fultineer, Jeremie Kalfon, **Javad Noorbakhsh**, Isabella Boyle, et al. *Nature Communications*, 2024.
6. [The enzyme glutamate-cysteine ligase \(GCL\) is a target for ferroptosis induction in cancer](#). John K Eaton, Priya Chatterji, Laura Furst, Sneha Basak, **Javad Noorbakhsh**, et al. *bioRxiv*, 2024.
7. [High-throughput deconvolution of 3D organoid dynamics at cellular resolution for cancer pharmacology with Cellos](#). Patience Mukashyaka, Pooja Kumar, David J Mellert, Shadae Nicholas, **Javad Noorbakhsh**, et al. *Nature Communications*, 2023.
8. [Computational estimation of quality and clinical relevance of cancer cell lines](#). Lucia Trastulla, **Javad Noorbakhsh**, Francisca Vazquez, James McFarland, Francesco Iorio. *Molecular Systems Biology*, 2022.
9. [Bridging the gap between cancer cell line models and tumours using gene expression data](#). **Javad Noorbakhsh**, Francisca Vazquez, James M McFarland. *British Journal of Cancer*, 2021.

10. [Deep learning-based cross-classifications reveal conserved spatial behaviors within tumor histological images.](#) **Javad Noorbakhsh**, Saman Farahmand, Sandeep Namburi, Dennis Caruana, David Rimm, et al. *Nature Communications*, 2020.
11. [Repeat expansions confer WRN dependence in microsatellite-unstable cancers.](#) Niek van Wietmarschen, Sriram Sridharan, William J Nathan, Anthony Tubbs, **Javad Noorbakhsh**, et al. *Nature*, 2020.
12. [Treating cancer as an invasive species.](#) **Javad Noorbakhsh**, Zi-Ming Zhao, James C Russell, Jeffrey H Chuang. *Molecular Cancer Research*, 2020.
13. [Integrative deep learning for PanCancer molecular subtype classification using histopathological images and RNAseq data.](#) Fatima Zare, **Javad Noorbakhsh**, Tianyu Wang, Jeffrey H Chuang, Sheida Nabavi. *ACM-BCB 2020*, 2020.
14. [Longitudinal molecular trajectories of diffuse glioma in adults.](#) Floris P Barthel, Kevin C Johnson, Frederick S Varn, Anzhela D Moskalik, **Javad Noorbakhsh**, et al. *Nature*, 2019.
15. [CCNE1 amplification is associated with poor prognosis in patients with triple negative breast cancer.](#) Zi-Ming Zhao, Susan E Yost, Katherine E Hutchinson, Sierra Min Li, **Javad Noorbakhsh**, et al. *BMC Cancer*, 2019.
16. [Machine learning in biology and medicine.](#) **Javad Noorbakhsh**, Harshpreet Chandok, R Krishna Murthy Karuturi, Joshy George. *Advances in Molecular Pathology*, 2019.
17. [Distribution-based measures of tumor heterogeneity are sensitive to mutation calling and lack strong clinical predictive power.](#) **Javad Noorbakhsh**, Hyunsoo Kim, Sandeep Namburi, Jeffrey H Chuang. *Scientific Reports*, 2018.
18. [High-resolution deconstruction of evolution induced by chemotherapy treatments in breast cancer xenografts.](#) Hyunsoo Kim, Pooja Kumar, Francesca Menghi, **Javad Noorbakhsh**, Eliza Cerveira, et al. *Scientific Reports*, 2018.
19. [Uncertainties in tumor allele frequencies limit power to infer evolutionary pressures.](#) **Javad Noorbakhsh**, Jeffrey H Chuang. *Nature Genetics*, 2017.
20. [From intracellular signaling to population oscillations: bridging size-and time-scales in collective behavior.](#) Allyson E Sgro, David J Schwab, **Javad Noorbakhsh**, Troy Mestler, Pankaj Mehta, et al. *Molecular Systems Biology*, 2015.
21. [Modeling oscillations and spiral waves in Dictyostelium populations.](#) **Javad Noorbakhsh**, David J Schwab, Allyson E Sgro, Thomas Gregor, Pankaj Mehta. *Physical Review E*, 2015.
22. [Two-dimensionality of yeast colony expansion accompanied by pattern formation.](#) Lin Chen, **Javad Noorbakhsh**, Rhys M Adams, Joseph Samaniego-Evans, Germaine Agollah, et al. *PLoS Computational Biology*, 2014.
23. [Intrinsic noise of microRNA-regulated genes and the ceRNA hypothesis.](#) **Javad Noorbakhsh**, Alex H Lang, Pankaj Mehta. *PLoS One*, 2013.

PATENT

[Convolutional neural networks for classification of cancer histological images.](#) Jeffrey Hsu-Min Chuang, **Javad Noorbakhsh**, Kourosh Zarringhalam, Saman Farahmand, Mohammad Soltanieh-ha. *U.S. Patent Application US 2025/0378559 A1*, 2025.